

PHISHGUARD AI

Anirudh S (Roll Number: 23035010660)

Term Project Report Trimester VIII

B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
anirudh.s@op.iitg.ac.in

ABSTRACT

Email phishing attacks remain one of the most common cybersecurity threats faced by individuals and organizations. Fraudulent emails often attempt to deceive users into revealing sensitive information such as passwords, banking credentials, and personal data. Manual identification of such messages becomes increasingly difficult due to the growing volume and sophistication of phishing campaigns.

This project presents PhishGuard-AI, a machine learning based phishing detection system designed to classify suspicious email content automatically. The proposed system combines Natural Language Processing (NLP), machine learning classification, and rule-based phishing pattern analysis to improve detection reliability. Textual content is first preprocessed through normalization and stop-word removal. TF-IDF vectorization is then applied to convert textual information into numerical features suitable for machine learning. A Linear Support Vector Classifier (LinearSVC) is trained on a dataset containing approximately five thousand labeled spam and legitimate messages. In addition to machine learning predictions, a phishing keyword analysis module evaluates the presence of suspicious patterns frequently observed in phishing attacks.

Experimental evaluation demonstrates strong classification performance with an accuracy of 98.21%, precision of 98.51%, recall of 88.00%, and F1-score of 92.96%. The developed Streamlit application provides an intuitive interface for real-time phishing detection and risk assessment.

Index Terms— Machine Learning, Natural Language Processing, Email Security, Phishing Detection, LinearSVC, Streamlit

1. INTRODUCTION

Electronic communication has become an essential component of modern personal and professional activities. However, the widespread use of email has also resulted in an increase in phishing attacks aimed at stealing confidential information. Cybercriminals frequently impersonate trusted organizations and encourage users to click malicious links, verify credentials, or disclose sensitive information.

Traditional spam filtering techniques provide partial protection but may fail to identify sophisticated phishing attempts that use persuasive language and social engineering tactics. Consequently, automated approaches based on machine learning have gained significant attention due to their ability to learn patterns from historical data and generalize to unseen messages.

The objective of this project is to design and implement a lightweight phishing detection system capable of identifying suspicious email content automatically. The proposed solution combines Natural Language Processing techniques, TF-IDF feature extraction, and a Linear Support Vector Classifier to classify messages as spam or legitimate. To further strengthen detection capability, a rule-based phishing keyword analyzer is integrated into the final decision-making process.

The developed system is deployed through a Streamlit web interface that enables users to submit email content and receive an immediate risk assessment. The project demonstrates the practical application of machine learning techniques in cybersecurity while maintaining computational efficiency and ease of deployment.

The remainder of this report is organized as follows. Section 2 presents the problem statement and objectives. Section 3 discusses the methodology and implementation details. Experimental evaluation is described in Section 4. Sections 5 and 6 present discussion, conclusions, and future directions.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Phishing emails represent a major cybersecurity challenge because they are specifically designed to exploit human trust and encourage unsafe actions. Attackers commonly use urgent language, account verification requests, password reset notifications, and fraudulent reward schemes to manipulate users.

Manual inspection of every incoming email is impractical in real-world environments where large volumes of messages are received daily. Therefore, an automated system capable of identifying suspicious content is required. The challenge lies in accurately distinguishing legitimate messages from potentially harmful communications while maintaining low computational complexity and high prediction accuracy.

This project addresses the problem by developing a machine learning based phishing detection system that analyzes email content and classifies messages according to their likelihood of being malicious.

2.2. Objectives

The primary objectives of the project are:

1. To study phishing attacks and common indicators present in malicious emails.

2. To preprocess textual email content using Natural Language Processing techniques.
3. To convert textual information into numerical representations using TF-IDF vectorization.
4. To train and evaluate a Linear Support Vector Classifier for spam classification.
5. To integrate rule-based phishing keyword detection for enhanced decision making.
6. To develop an interactive Streamlit application for real-time phishing analysis.
7. To evaluate system performance using standard machine learning metrics.

3. METHODOLOGY / APPROACH

3.1. Overall Workflow

Figure 1 illustrates the workflow of the proposed phishing detection system.

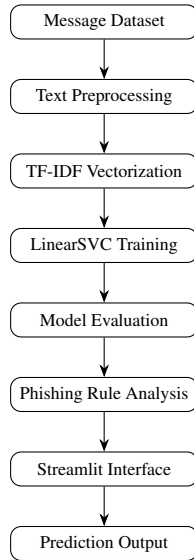


Fig. 1. Workflow of the PhishGuard-AI system

The workflow begins with loading a labeled collection of legitimate and spam messages. Each message undergoes preprocessing operations including lowercasing, removal of non-alphabetic characters, and elimination of common stop words. The cleaned text is transformed into TF-IDF vectors which serve as input features for machine learning.

The LinearSVC classifier is trained using these vector representations and subsequently evaluated on unseen samples. The trained model is saved and integrated into the Streamlit application. During prediction, phishing keyword analysis is additionally performed to identify suspicious patterns commonly associated with fraudulent messages.

3.2. Technical Details

The preprocessing module was implemented using Python and the Natural Language Toolkit (NLTK). Text normalization converts all characters to lowercase and removes irrelevant symbols. English

stop words are eliminated to reduce noise and improve feature quality.

TF-IDF vectorization was applied with a maximum vocabulary size of 5000 features and support for both unigrams and bigrams. This representation emphasizes informative terms while reducing the influence of commonly occurring words.

The classification model employed in this project is the Linear Support Vector Classifier (LinearSVC). Support Vector Machines are widely used in text classification tasks because of their effectiveness in high-dimensional feature spaces.

The optimization objective of the classifier can be represented as:

$$\min \frac{1}{2} ||w||^2 + C \sum_{i=1}^N \xi_i \quad (1)$$

where w denotes the separating hyperplane, C represents the regularization parameter, and ξ_i denotes classification errors.

In addition to machine learning predictions, the application evaluates the occurrence of phishing-related keywords such as account verification requests, password reset instructions, urgent actions, banking references, and login confirmations. Messages exhibiting multiple suspicious indicators are assigned elevated risk levels.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

The phishing detection system was implemented using Python and evaluated on a labeled dataset containing approximately 5,000 messages belonging to spam and legitimate categories. The dataset was divided into training and testing subsets using an 80:20 split to evaluate generalization performance.

The development environment consisted of Python, Pandas, NLTK, Scikit-learn, and Streamlit. Text preprocessing was performed prior to feature extraction, and TF-IDF vectorization was configured with a vocabulary size of 5000 features and support for unigram and bigram representations.

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score

These metrics provide a comprehensive assessment of classification performance, particularly for cybersecurity applications where false positives and false negatives have practical implications.

4.2. Results

The trained LinearSVC model achieved strong classification performance on the test dataset. The obtained evaluation metrics are presented in Table 1.

Table 1. Performance Evaluation Results

Metric	Value
Accuracy	98.21%
Precision	98.51%
Recall	88.00%
F1-Score	92.96%

The high accuracy indicates that the classifier successfully distinguishes between spam and legitimate messages. Precision exceeding 98% demonstrates that most messages classified as malicious are indeed suspicious. The achieved F1-score reflects a strong balance between precision and recall.

The Streamlit application successfully integrated machine learning predictions with phishing keyword analysis and provided risk assessments categorized as Low, Medium, High, and Very High. The system generated predictions in real time while maintaining an intuitive user interface.

5. DISCUSSION

The experimental results demonstrate that machine learning techniques can effectively identify suspicious textual patterns associated with phishing and spam messages. The combination of TF-IDF vectorization and LinearSVC produced strong classification performance while maintaining computational efficiency.

One important advantage of the proposed approach is its simplicity. Unlike deep learning models that often require substantial computational resources and large datasets, the implemented classifier can be trained and deployed with relatively modest hardware requirements. This makes the system suitable for educational and small-scale security applications.

The integration of phishing keyword analysis further improves interpretability by highlighting suspicious indicators present within submitted content. Rather than relying solely on machine learning predictions, the system incorporates domain-specific rules that provide additional context regarding potential threats.

Despite encouraging results, several limitations remain. The current model primarily analyzes textual content and does not evaluate URLs, attachments, sender reputation, or metadata that may also contribute to phishing detection. Furthermore, attackers continuously modify phishing strategies, requiring periodic retraining and rule updates to maintain effectiveness.

Overall, the project successfully demonstrates the practical application of Natural Language Processing and machine learning techniques within cybersecurity.

6. CONCLUSION AND FUTURE WORK

This project presented the design and implementation of PhishGuard-AI, a machine learning based phishing detection system developed to identify suspicious email content automatically. The solution combines Natural Language Processing, TF-IDF feature extraction, LinearSVC classification, and phishing keyword analysis to provide reliable threat assessment.

Experimental evaluation produced an accuracy of 98.21%, precision of 98.51%, recall of 88.00%, and an F1-score of 92.96%, indicating strong classification capability. The Streamlit-based interface enables real-time interaction and demonstrates how machine learning can be applied to practical cybersecurity problems.

Future enhancements may include the integration of dedicated phishing email datasets, URL analysis, attachment inspection, sender verification mechanisms, and advanced deep learning models such as recurrent neural networks or transformer-based architectures. Additional improvements could also include cloud deployment, user authentication, and continuous model retraining pipelines.

7. ARTIFACTS AND DEMONSTRATIONS

The following artifacts are associated with the project:

- **GitHub Repository:**<https://github.com/an1rudhks/PhishGuard-AI>
- **Video Demonstration:**<https://youtu.be/4jvQiw581Y4>

The repository contains source code, dataset files, frontend implementation, and supporting documentation as required.

8. REFERENCES

- [1] S. Raschka and V. Mirjalili, *Machine Learning with Python*, Packt Publishing, 2019.
- [2] Scikit-learn Developers, *Scikit-learn Documentation*, Available: <https://scikit-learn.org>
- [3] Streamlit Documentation, Available: <https://streamlit.io>